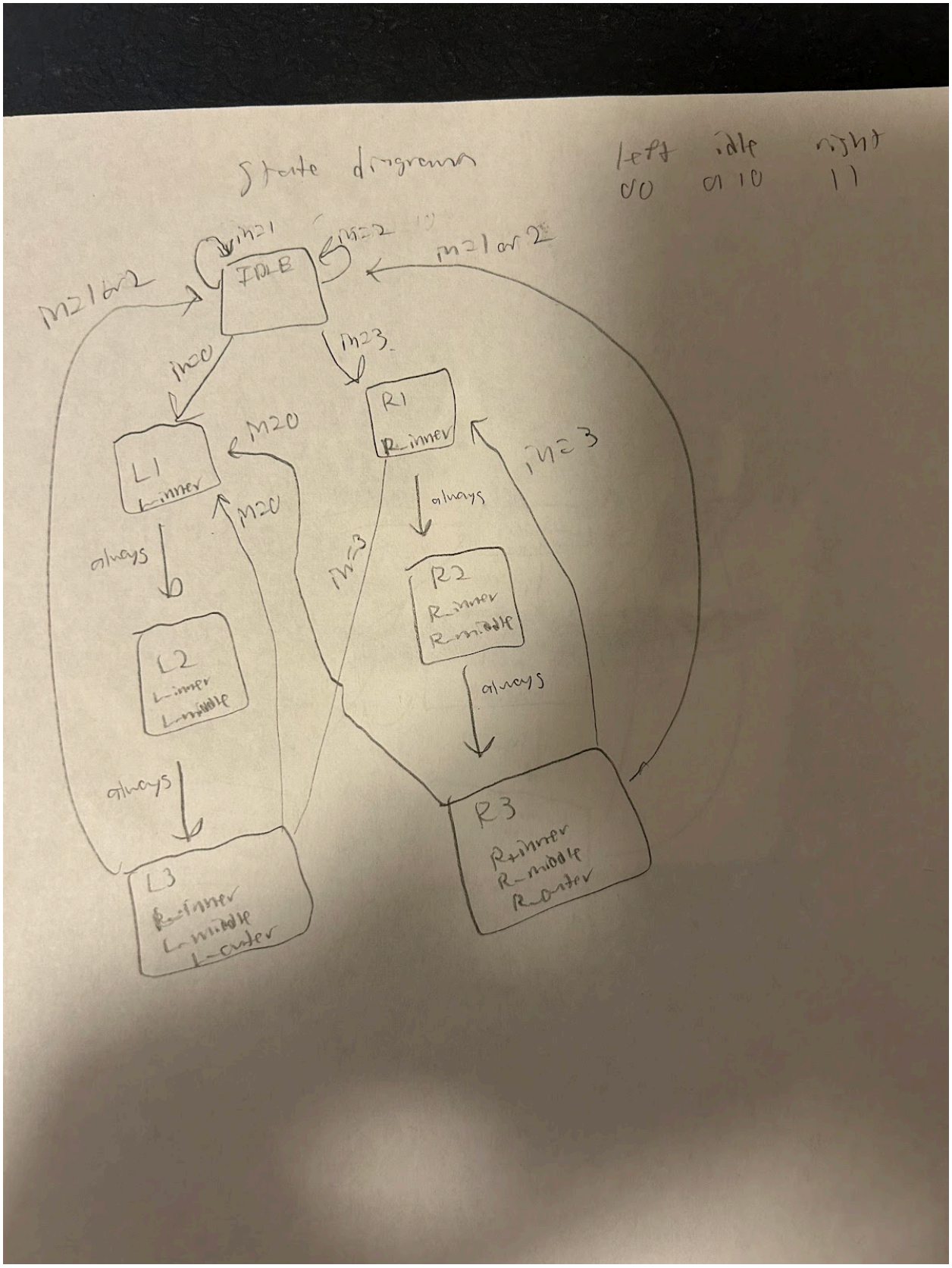
Dylan Thornburg & Ryan Kinris ECEN 127 Lab 4 Prelab

(i) Given the description above, particularly the discussion of the inputs and outputs, define the interface for a Verilog module. This will require you to give explicit names to signals. If you define any of the signals as buses, make sure the width is clear.

```
module turn_signal(
        input clk, rst,
        input [1:0] signal,
        output reg L_Inner, L_Middle, L_Outer, R_Inner, R_Middle, R_Outer
);
```

(ii) Create a state diagram that maps out the state transitions. Make sure your transitions are clearly indicated with the input conditions, and that your states show what output signals are being asserted.



I forgot to put this in, but if rst is asserted, we always go back to IDLE. Regardless of

lightstate. When asserted, it will not complete the full three light cycle but that wasnt a requirement (just something we could assume) so we should be fine.

(iii) Outline a handful of important test cases that you will run against your design to make sure it's working properly.

We will test all the major functions to make sure they work as intended, then we will see if they still function when we change from a signal to idle or another signal before the first one finishes to see if the lights fully illuminate (if switching to idle it shouldnt, if other signal, it should).

Note that having the state machine coded in Verilog is not a requirement for pre-lab. It would be a good idea to start typing in your code before lab.
OK